# STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-CURVE-ISOLATED

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 3030-3052 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 3054-3070, the complete author proof (15 body lines).

*Editorial note.* Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 36 of queue_examples-defos.json (pb=729, stmt 3030-3052, proof 3054-3070).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0DY7: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1425–1440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0DYX: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1600–1608. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0DZN: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2971–2988. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0DZF: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2874–2890. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01ZY: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3070–3070. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0A25: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3070–3070. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *In Example 0DY7 let $X$ be a scheme over $k$. Assume*

(1) *$X$ is separated, finite type over $k$ and $\dim(X) \leq 1$,*
(2) *$X \to \mathrm{Spec}(k)$ is smooth except at the closed points $p_1, \ldots, p_n \in X$.*

*Let $\mathcal{O}_{X,p_1}$, $\mathcal{O}^h_{X,p_1}$, $\mathcal{O}^\wedge_{X,p_1}$ be the local ring, henselization, completion. Consider the maps of deformation categories*

$$\mathcal{D}ef_X \longrightarrow \prod \mathcal{D}ef_{\mathcal{O}_{X,p_i}} \longrightarrow \prod \mathcal{D}ef_{\mathcal{O}^h_{X,p_i}} \longrightarrow \prod \mathcal{D}ef_{\mathcal{O}^\wedge_{X,p_i}}$$

*The first arrow is smooth and the second and third arrows are smooth and induce isomorphisms on tangent spaces.*

*Proof.* Choose an affine open $U_2 \subset X$ containing $p_1, \ldots, p_n$ and the generic point of every irreducible component of $X$. This is possible by Varieties, Lemma 0A25 and Properties, Lemma 01ZY. Then $X \setminus U_2$ is finite and we can choose an affine open $U_1 \subset X \setminus \{p_1, \ldots, p_n\}$ such that $X = U_1 \cup U_2$. Set $U_{12} = U_1 \cap U_2$. Then $U_1$ and $U_{12}$ are smooth affine schemes over $k$. We conclude that

$$\mathcal{D}ef_X \longrightarrow \mathcal{D}ef_{U_2}$$

is smooth by Lemma 0DZN. Applying Lemmas 0DYX and 0DZF we win. □